SIMULATION & ANALYSIS

EX26-0002 · plastic_cover

Rev A · plastic · opening direction Z · wall 2.50 mm · 2026-10-26 · analytical estimate, not a flow simulation

Uses example rates – not for negotiation

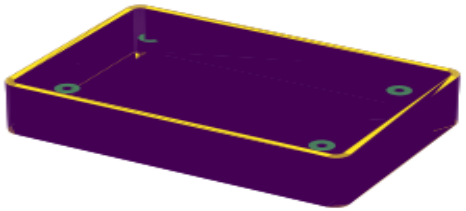
Simulation rules are EXAMPLE values.

Corrections required

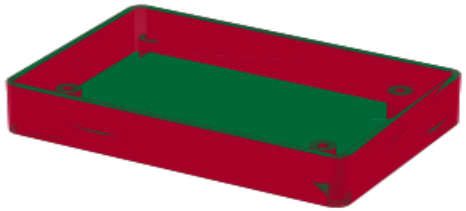
- Add draft of at least 1° on 248.4 cm² of walls along the opening direction Z.
- Core out thick sections: 15.1 cm² is thicker than 1.6 × the 2.50 mm wall (sink / porosity risk).
- Estimated cycle 1774.3 s is longer than the planned 25 s (Parameters → cycle_time_s).

Maps

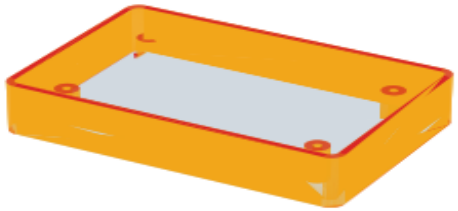
Wall thickness 2.50–30.00 mm (yellow = thick)



Draft to Z (red = below 1°)



Risk: red sink, blue thin, orange low draft, purple undercut



Results

	Value
Low draft area (cm²)	248.4
Sink-risk area (cm²)	15.1
Thin-wall area (cm²)	0.0
Undercut area (cm²)	0.0

	Value
Cycle time (s)	1774.3
Clamp force (t)	24.8
Machine	IMM-80 (80 t)

Calculation steps (Mould & die database → Simulation rules)

Step	Formula	Value	Unit
Minimum draft on walls along the opening direction	1	1.000	°
Thin-wall flag below this × nominal wall	0.5	0.500	
Sink-risk flag above this × nominal wall	1.6	1.600	
Melt temperature	230	230.000	°C
Mould temperature	50	50.000	°C
Ejection temperature	90	90.000	°C
Thermal diffusivity of the plastic	0.09	0.090	mm²/s
Cooling time of the thickest wall (centre-line formula)	max_wall_mm ** 2 / (9.8696 * diffusivity_mm2_s) * log(4 / 3.1416 * (melt_c - mould_c) / (eject_c - mould_c))	1,768.703	s
Injection and packing time	0.5 + shot_weight_g / 50	1.618	s
Mould open, eject, close (+ side actions)	4 + (2 if slides + lifters > 0 else 0)	4.000	s
1 = vertical machine with shuttle / rotary table (Insert moulding rules)	1 if inserts_per_shot > vertical_machine_if_inserts_per_shot_over else 0	0.000	
Insert loading time added to the cycle	inserts_per_shot * manual_load_time_s_per_insert * (1 - use_table * table_hidden_load_pct / 100)	0.000	s
Cycle time	fill_s + cooling_s + open_eject_s + insert_load_s	1,774.322	s
Flow length ÷ wall	flow_length_mm / max(wall_mm, 0.1)	60.000	
Injection pressure needed	min(1800, 300 + 4 * flow_ratio)	540.000	bar
Mean cavity pressure	injection_bar * 0.4	216.000	bar
Clamp force	projected_area_cm2 * cavities * cavity_bar * clamp_safety_factor / 1000	24.827	t